

ELEC 221 Lecture 08

DT Fourier series and filters

Tuesday 6 October 2026

Announcements

- Quiz 4 today
- Tutorial Assignment 2 due Friday 23:59

$$X_1(t) = 2 + 3 \sin(2\omega t) + 2 \cos(3\omega t)$$

$$X_2(t) = 1 + 3 \sin(\omega t) + 2 \cos(\omega t)$$

$$X_1(t) + 2X_2(t) = \sum_k C_k e^{jk\omega t}$$

$C_0, C_1, \dots, C_n \Rightarrow n?$
 $\underbrace{C_0, C_1, C_2, C_3}_{\text{will have imag.}}$ will appear
real *real* *real*

$\Rightarrow$ sin one have j , cos ones real.

Add them together:

$$X_1(t) + 2X_2(t) = 4 + 4 \cos(\omega t) + 9 \sin(2\omega t) + 2 \cos(3\omega t)$$

$$C_0 = 4$$

$$C_1 = 2$$

$$C_2 = \frac{9}{2j} = -\frac{9}{2}j$$

$$C_3 = 1$$

$$4 \cos(\omega t) = 4 \cdot \left(\frac{e^{j\omega t} + e^{-j\omega t}}{2} \right) = 2 \cdot e^{j\omega t} + 2 \cdot e^{-j\omega t}$$

$$9 \cdot \frac{e^{2j\omega t} - e^{-2j\omega t}}{2j}$$

$$C_{-1} = C_1^*$$

$$C_{-2} = C_2^*$$

$$2 \sin(\omega t) + \cos(\omega t)$$

$$2 \cdot \frac{e^{j\omega t} - e^{-j\omega t}}{2j} + \frac{e^{j\omega t} + e^{-j\omega t}}{2}$$

$$= \left(\frac{1}{j} + \frac{1}{2} \right) e^{j\omega t} + \left(\frac{1}{2} - \frac{1}{j} \right) e^{-j\omega t}$$

$$C_1 = \frac{1}{2} + \frac{1}{j}$$

$$C_{-1} = C_1^* = \frac{1}{2} - \frac{1}{j}$$

$$C_1 = \frac{\omega}{2\pi} \int_0^{\frac{\omega}{2\pi}} e^{-j\omega t} (2 \sin \omega t + \cos \omega t) dt$$

$$x(t) = 2t + 1$$

Last time

Fourier synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega t}$$

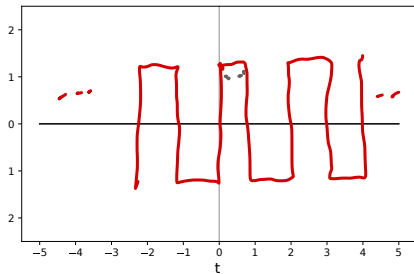
Fourier analysis equation:

$$C_k = \frac{1}{T} \int_T e^{-jk\omega t} x(t) dt$$

Last time

We evaluated the Fourier series coefficients of a square wave:

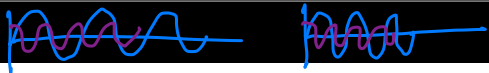
$$x(t) = \begin{cases} 1, & 0 \leq t < \pi, \\ -1, & \pi \leq t < 2\pi \end{cases}$$



*finite sum:
Gibbs
phenomenon.*

$$x(t) = \sum_{k=1}^{\infty} b_k \sin(kt), \quad b_k = \begin{cases} 0, & k \text{ is even} \\ \underline{4/k\pi}, & k \text{ is odd} \end{cases}$$

Last time



We determined how Fourier series coefficients transform. $x(t) + x(t-1)$

Superposition of two signals with same ω :

$$x(t) = \sum_k a_k e^{jk\omega t} \quad y(t) = \sum_k b_k e^{jk\omega t}$$

$$\Rightarrow Ax(t) + By(t) = \sum_k (Aa_k + Bb_k) e^{jk\omega t}$$

Time shift

$$x(t) = \sum_k c_k e^{jk\omega t} \quad x(t-t_0) = \sum_k \underbrace{c_k e^{-jk\omega t_0}}_{\text{new coeff.}} e^{jk\omega t}$$

Time scale

$$x(\alpha t) \Rightarrow \sum_k c_k e^{jk(\alpha\omega)t}$$

Multiplication leads to convolution:

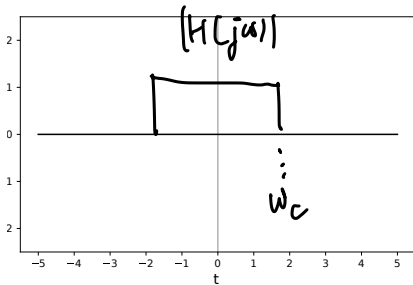
$$z(t) = x(t) y(t)$$

$$\sum_k c_k e^{jk\omega t} \quad c_k = \sum_{l=-\infty}^{\infty} a_l b_{k-l}$$

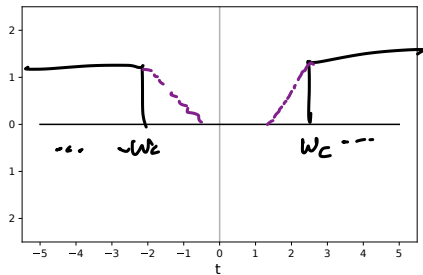
Last time

We used the frequency response to define some simple filters.

$$H(j\omega) = \begin{cases} 1 & |\omega| < \omega_c \\ 0 & |\omega| > \omega_c \end{cases}$$



$$H(j\omega) = \begin{cases} 1 & |\omega| > \omega_c \\ 0 & |\omega| < \omega_c \end{cases}$$



Today

Learning outcomes:

- Determine the frequency response of an RC filter circuit
- Compute the fundamental period and frequency of DT signals
- Evaluate Fourier series coefficients of DT signals
- Define and distinguish between finite-impulse-response (FIR) and infinite-impulse-response (IIR) filters in DT

Example: Lowpass filters in practice

Some filters made of physical components are described by differential equations.

Example: a lowpass filter using a resistor and a capacitor.

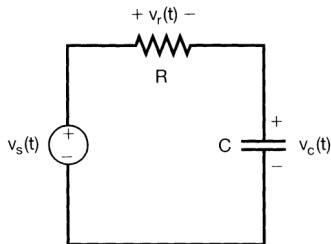
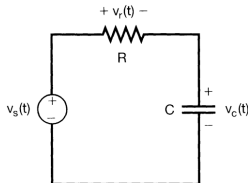


Image credit: Oppenheim chapter 3.10.

Example: lowpass filters in practice

What is the voltage across the capacitor if $v_s = e^{j\omega t}$?



Can derive two expressions for the current, using the resistor and capacitor respectively:

$$i(t) = \frac{v_s(t) - v_c(t)}{R}$$

$$i(t) = C \frac{dv_c(t)}{dt}$$

Example: lowpass filters in practice

$$A \cdot \frac{dy(t)}{dt} + y(t) = x(t)$$

Put these together to form a differential equation:

$$C \frac{dv_c(t)}{dt} = \frac{v_s(t) - v_c(t)}{R}$$

$$RC \frac{dv_c(t)}{dt} + v_c(t) = v_s(t)$$

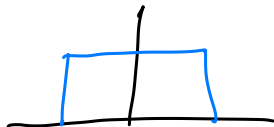
$$RC \cdot \frac{d}{dt} [H(j\omega) e^{j\omega t}] + H(j\omega) e^{j\omega t} = e^{j\omega t}$$

$$RC \cdot H(j\omega) \cdot j\omega e^{j\omega t} + H(j\omega) e^{j\omega t} = e^{j\omega t}$$

$$(RCj\omega + 1) H(j\omega) = 1$$

$$\Rightarrow H(j\omega) = \frac{1}{1 + j\omega RC}$$

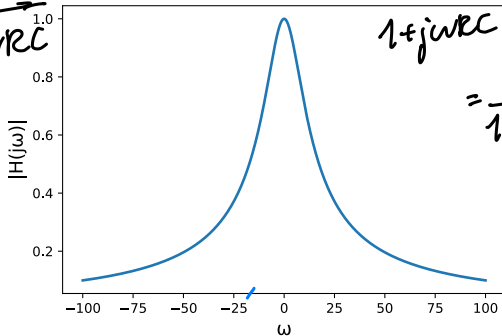
$$\begin{aligned} &v_s(t) = e^{j\omega t} \\ &v_c(t) \leftrightarrow V_c(t) \\ &V_c(t) = H(j\omega) \cdot V_s(t) \\ &= H(j\omega) e^{j\omega t} \end{aligned}$$



Example: lowpass filters in practice

Results in the following frequency response (setting $RC = 0.1$):

$$H(j\omega) = \frac{1}{1 + j\omega RC}$$



$$\begin{aligned} & \frac{1}{1 + j\omega RC} \cdot \frac{1 - j\omega RC}{1 - j\omega RC} \\ &= \frac{1}{1 + (\omega RC)^2} - j \frac{\omega RC}{1 + (\omega RC)^2} \end{aligned}$$

Adjusting the value of RC controls the frequency response. Increasing RC cuts off more frequencies.

DT complex exponential signals

Recall our CT representation of complex exponential signals:

$$x(t) = Ce^{\alpha t}$$

where α could be real or complex.

In DT, we write

$$x[n] = C\beta^n = C \cdot e^{\alpha n}$$

where β can be real or complex.

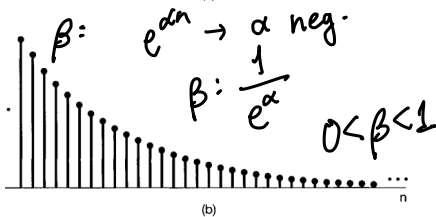
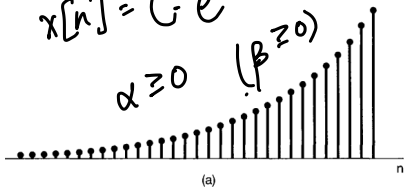
DT complex exponential signals

Case: $x[n] = C\beta^n$, where β is real.

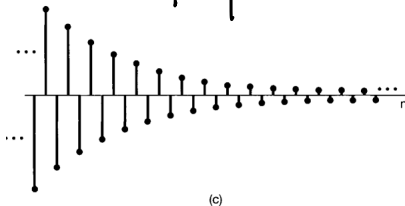
$$x[n] = C e^{\alpha \cdot n}$$

$$\alpha \geq 0$$

$$(\beta \geq 0)$$



$$-1 < \beta < 0$$



$$\beta < -1$$

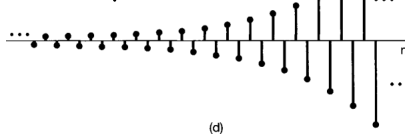
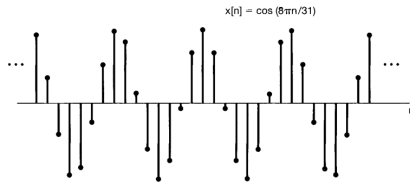
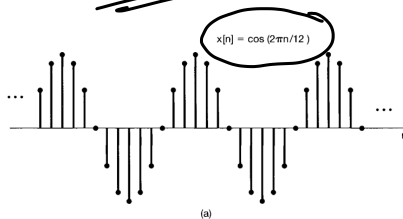


Image credit: Oppenheim chapter 1.3.

DT complex exponential signals

Case: $x[n] = C\beta^n$, where β is ~~any~~ ^{a complex number} ~~complex~~:

$$x[n] = e^{j\omega \cdot n} = \cos(\omega n) + j \sin(\omega n)$$



Frequency and period of DT complex exponential signals

While these might look similar to their CT counterparts, there is a **very important difference** relating to frequency.

In CT,

$$x(t) = e^{j\omega t} \sim \cos(\omega t) + j\sin(\omega t)$$

This is periodic with period $T = \frac{2\pi}{\omega}$

The bigger the frequency (ω) gets, the faster it oscillates!

Frequency and period of DT complex exponential signals

Exercise: consider the DT signal

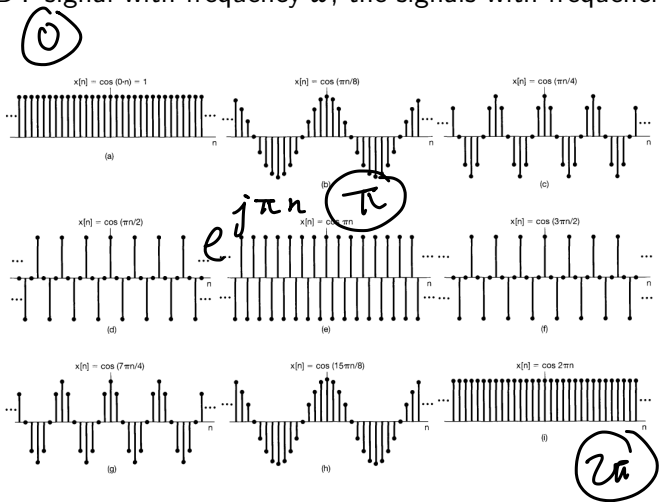
$$x[n] = e^{j\omega n} \rightarrow \text{integer.}$$

Does bigger ω always mean faster oscillation? If yes, why? If no, when does it stop getting faster?

$$\begin{aligned} e^{j(\omega + 2\pi)n} &= e^{j\omega n} e^{j \cdot 2\pi n} \\ &= e^{j\omega n} \underbrace{e^{j \cdot 2\pi n}}_1 \end{aligned}$$

Frequency and period of DT complex exponential signals

For a DT signal with frequency ω , the signals with frequencies



Frequency and period of DT complex exponential signals

Exercise: What are the fundamental periods of

$$x(t) = \cos(3t), \quad \text{and} \quad x[n] = \cos(3n)$$

try it (try plotting it!)
will start here Thursday

For next time

Content:

- The CT Fourier transform

Action items:

- ① Tutorial Assignment 2 due Friday 23:59

Recommended reading:

- From today's class: Oppenheim 3.5-3.12
- Suggested problems: 3.2, 3.10-3.12, 3.14, 3.16, 3.17, 3.19, 3.20 3.23, 3.27-3.32, 3.36, 3.39
- For next class: Oppenheim 4.1-4.3